

# Latex Testing

This document contains 10 mathematical formulas across algebra and calculus to test the LaTeX and MathML integration.

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## Algebra Formulas

### 1. Quadratic Formula

Used to find the roots of a quadratic equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

### 2. Difference of Squares

A standard algebraic identity for factoring polynomials:

$$a^2 - b^2 = (a - b)(a + b)$$

### 3. Binomial Theorem

Describes the algebraic expansion of powers of a binomial:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

### 4. Logarithm Change of Base

Used to convert logarithms from one base to another:

$$\log_a(x) = \frac{\log_b(x)}{\log_b(a)}$$

### 5. Slope-Intercept Equation

The standard equation of a straight line in Cartesian coordinates:

$$y = mx + b$$

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## Calculus Formulas

### 6. Limit Definition of the Derivative

Represents the instantaneous rate of change of a function:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

### 7. Fundamental Theorem of Calculus

Connects differentiation and integration:

$$\int_a^b f(x) dx = F(b) - F(a)$$

### 8. Taylor Series Expansion

Represents a function as an infinite sum of terms calculated from the values of its derivatives at a single point:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

### 9. Integration by Parts

A rule that transforms the integral of a product of functions:

$$\int u dv = uv - \int v du$$

### 10. Euler's Identity

A famous equation in complex analysis relating five fundamental mathematical constants:

$$e^{i\pi} + 1 = 0$$